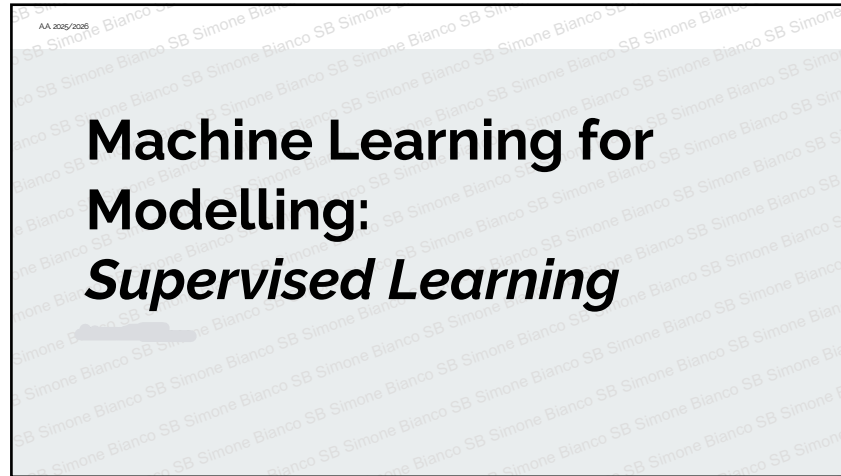
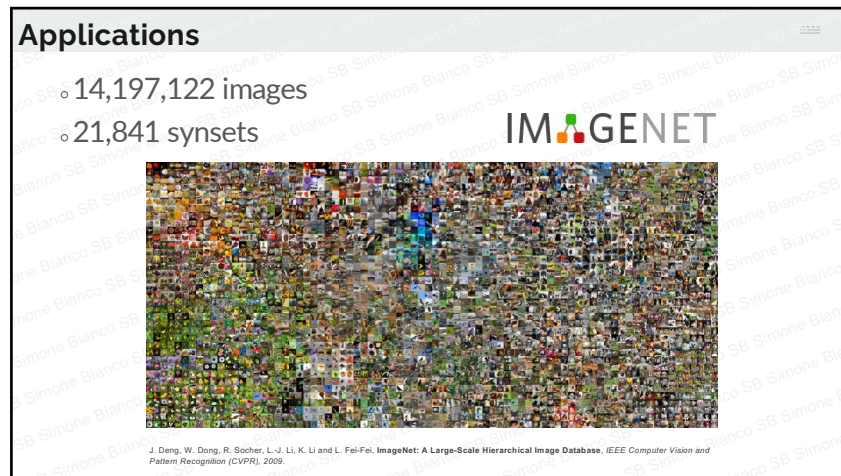




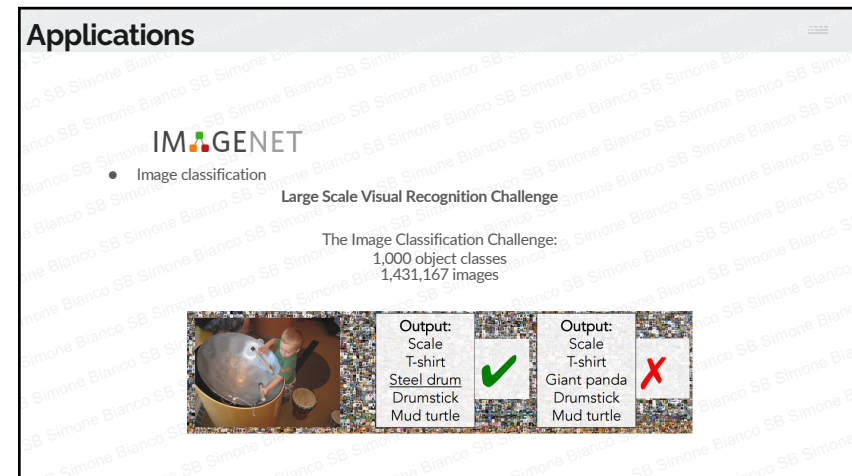
1



61



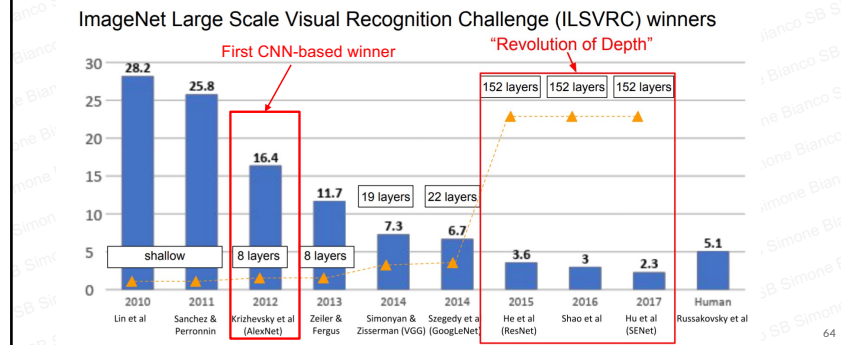
62



63

CNN Architectures

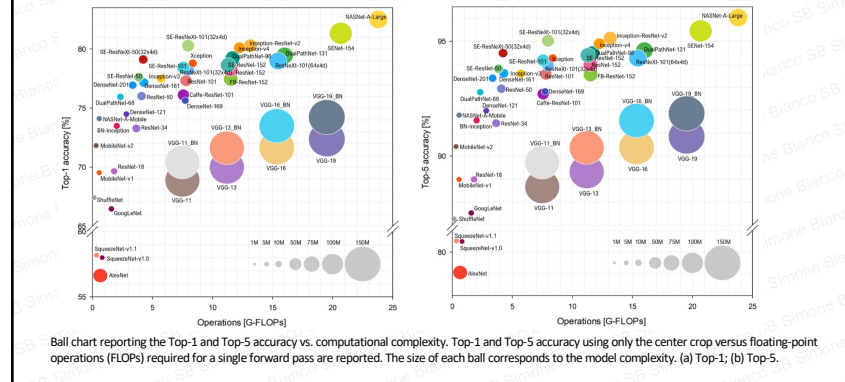
Imagenet classification error



64

CNN Architectures

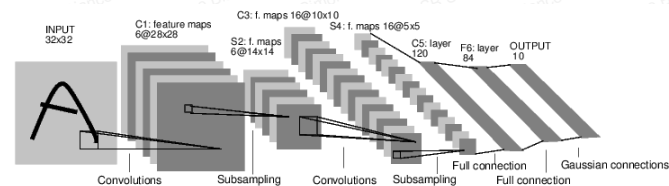
Imagenet classification error



65

CNN Architectures

LeNet (LeNet-5)

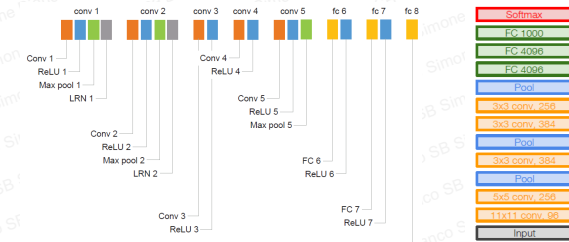


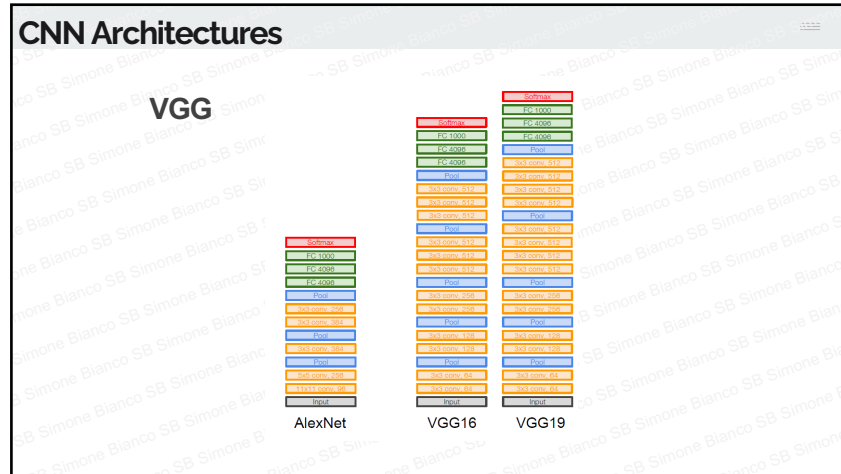
LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.

66

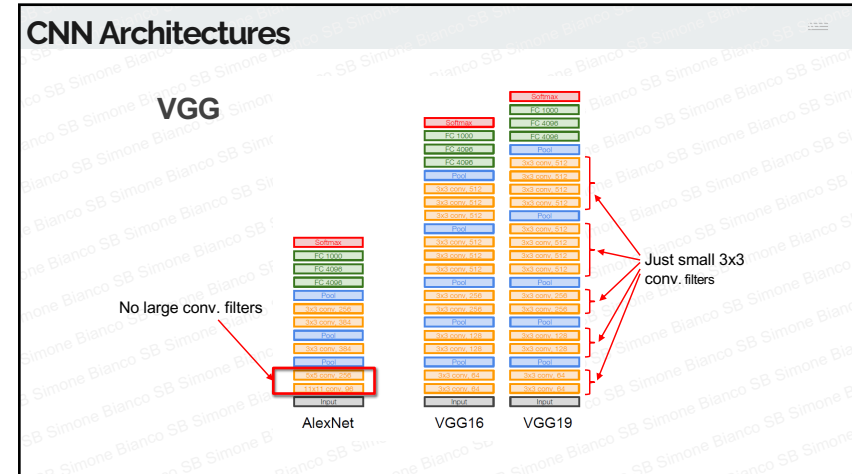
CNN Architectures

AlexNet

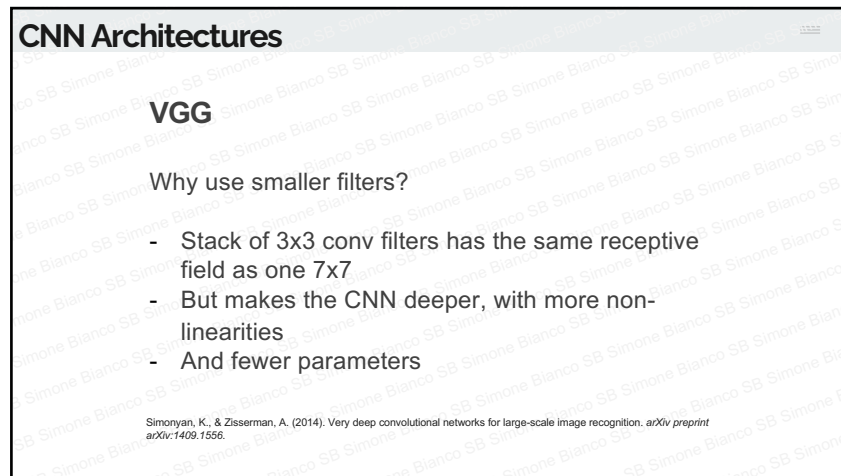




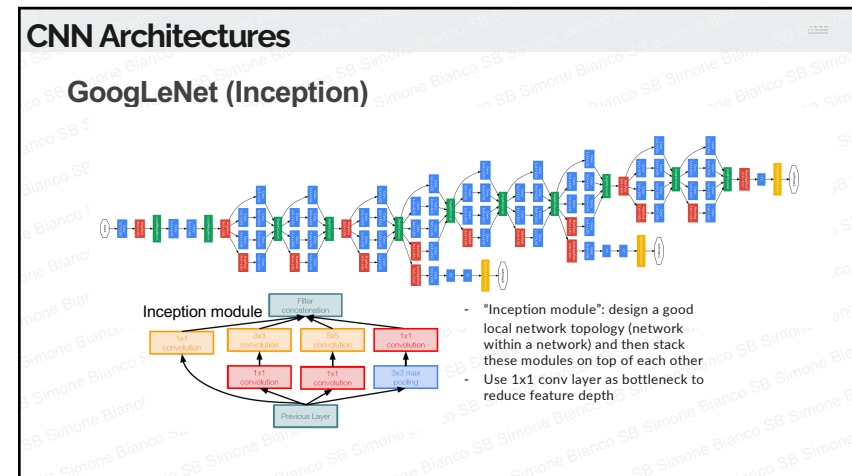
68



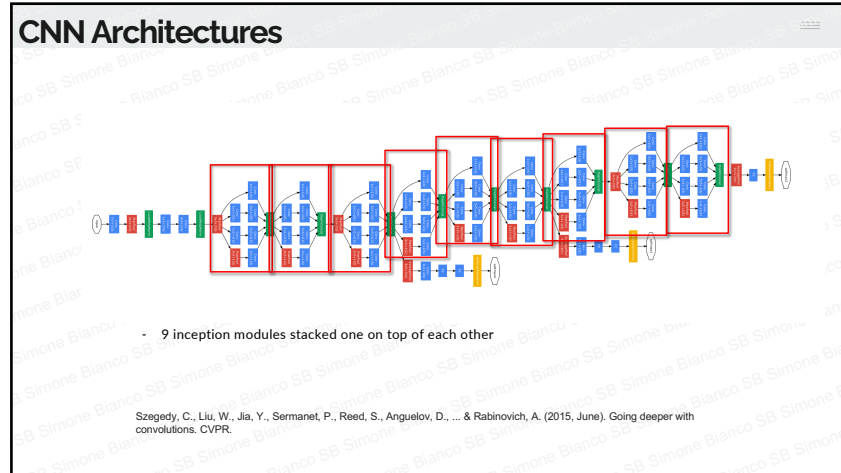
69



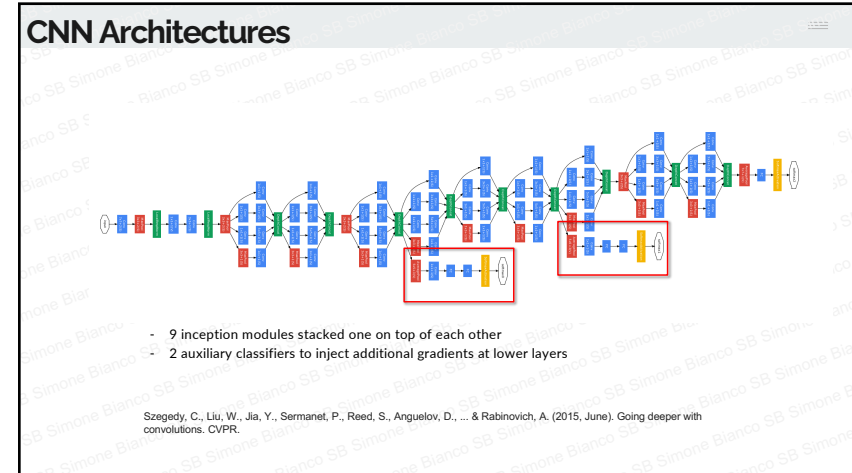
70



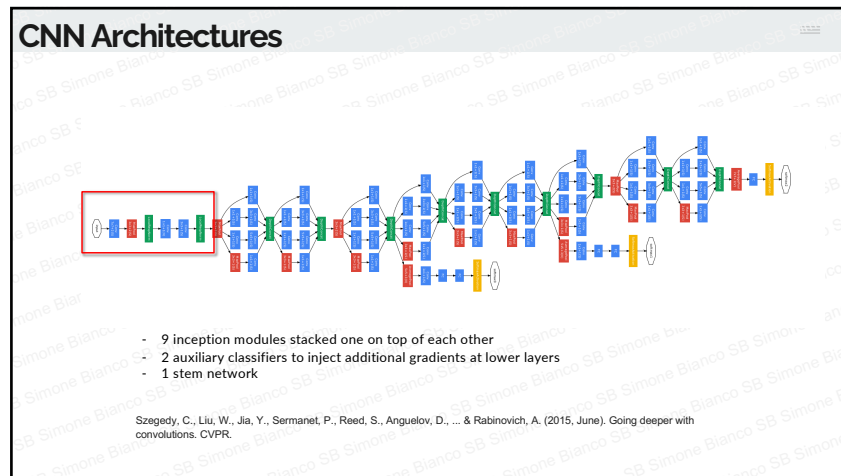
71



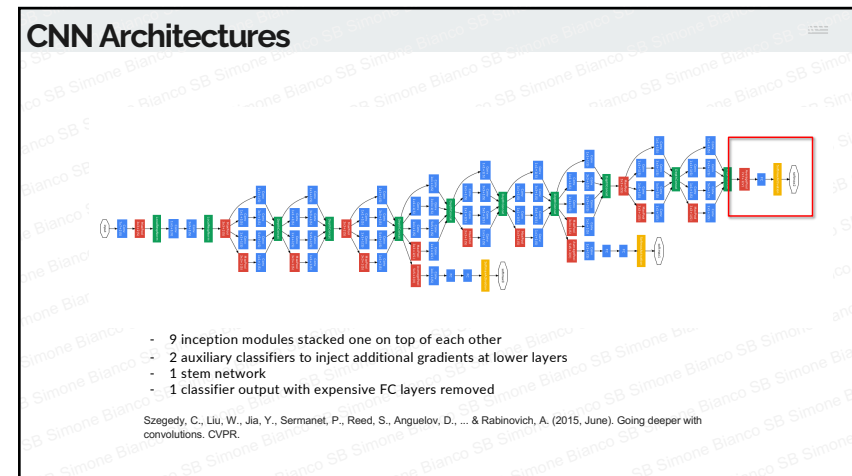
72



73



74

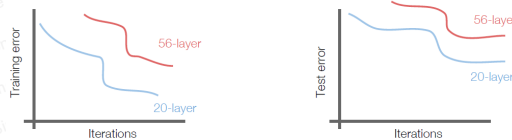


75

CNN Architectures

ResNet

- So far, the deeper the better
- Can we continue on adding layer and go deeper?



He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

76

CNN Architectures

ResNet

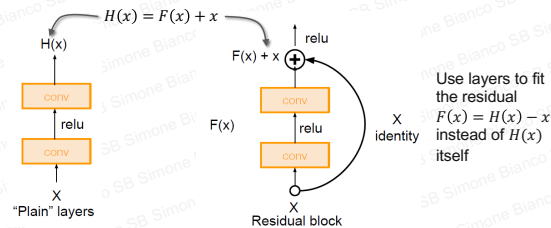
- So far, the deeper the better
- Can we continue on adding layer and go deeper?
- Unfortunately no
- Hypothesis: the problem is an optimization problem, deeper models are harder to optimize (e.g. vanishing gradients)

77

CNN Architectures

ResNet

Solution: Use network layers to fit a residual mapping instead of directly trying to fit a desired underlying mapping

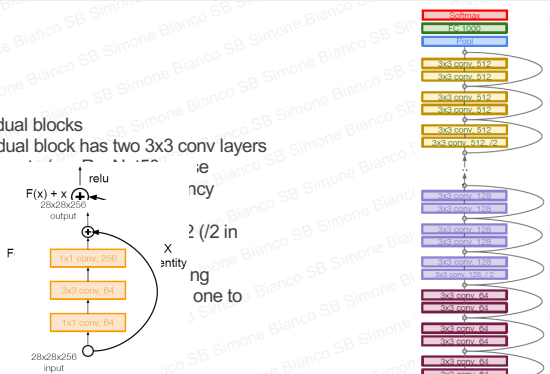


78

CNN Architectures

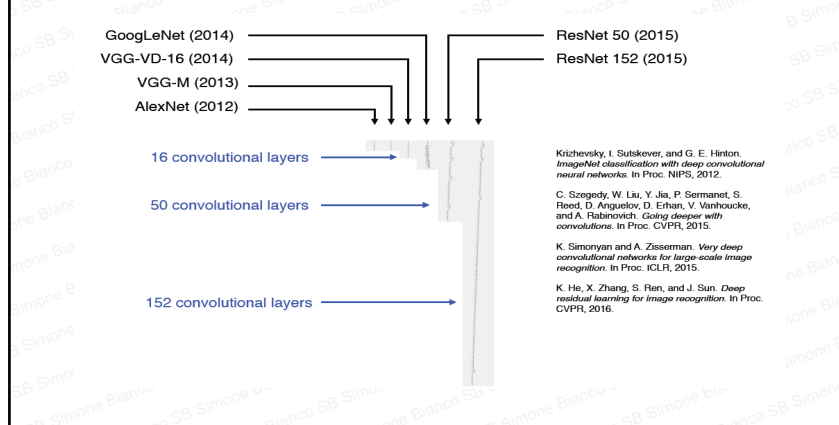
ResNet

- Stack residual blocks
- Every residual block has two 3x3 conv layers
- For deeper bottleneck
- Periodically downsample each time
- Additional No FC layer map to the



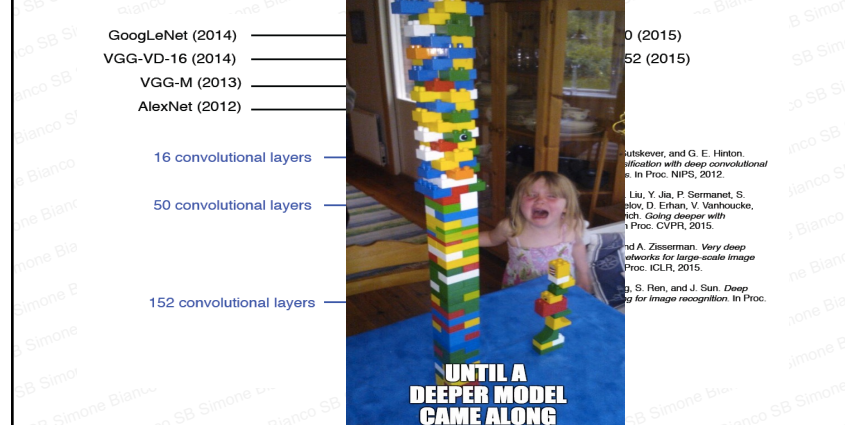
79

How deep is enough?



80

How deep is enough?



81

Transfer learning

- ## Transfer learning

- All the best performing architectures have millions of parameters to learn
- In order to tune them, very large databases are needed (e.g., rule of 10: 10x samples wrt number of parameters to train)



- What if we do not have such amount of data? Can we still use deep learning?

82

83

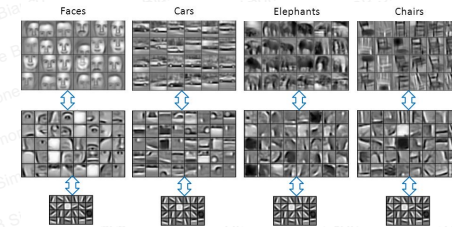
Transfer learning

- In practice, very few people train an entire CNN from scratch, because it is relatively rare to have a dataset of sufficient size.
- It is common to pretrain a CNN on a very large dataset (e.g., ImageNet which contains 1.2 million images with 1000 categories)
- and then use the CNN either for fine-tuning or as a fixed feature extractor for the task of interest.

84

Transfer learning

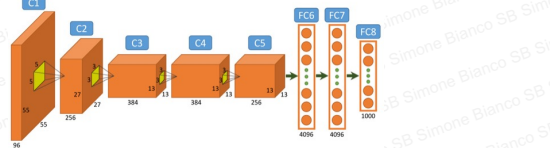
- Why does it work?
- Motivated by the observation that the earlier features of a CNN contain more generic features (e.g., edge detectors or color blob detectors) that should be useful to many tasks
- Later layers of the CNN becomes progressively more specific to the details of the classes contained in the original dataset.



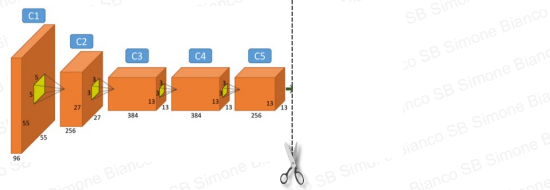
85

CNN fine-tuning

- Task 1 (e.g. ImageNet, 1000 categories)



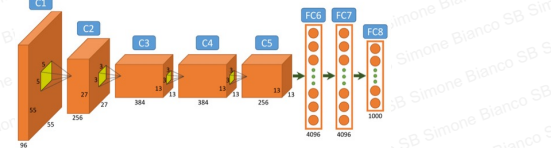
- Task 2 (10 categories)



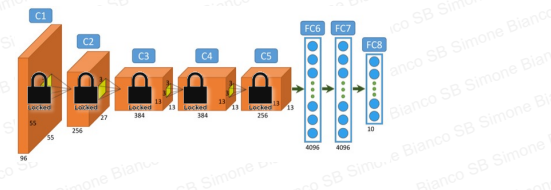
86

CNN fine-tuning

- Task 1 (e.g. ImageNet, 1000 categories)

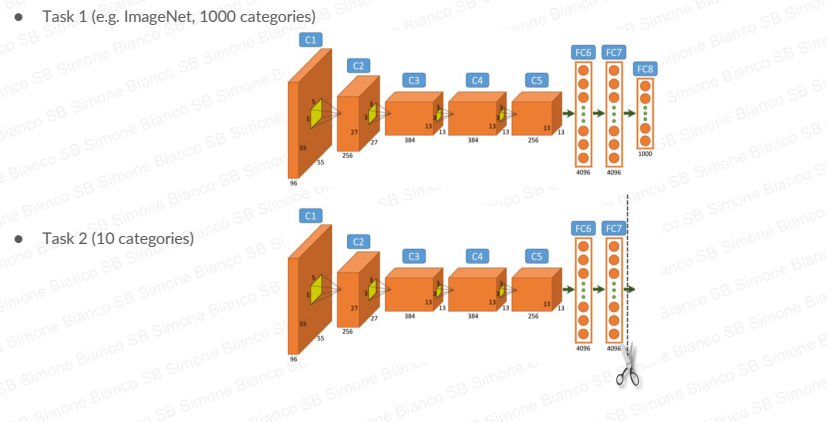


- Task 2 (10 categories)



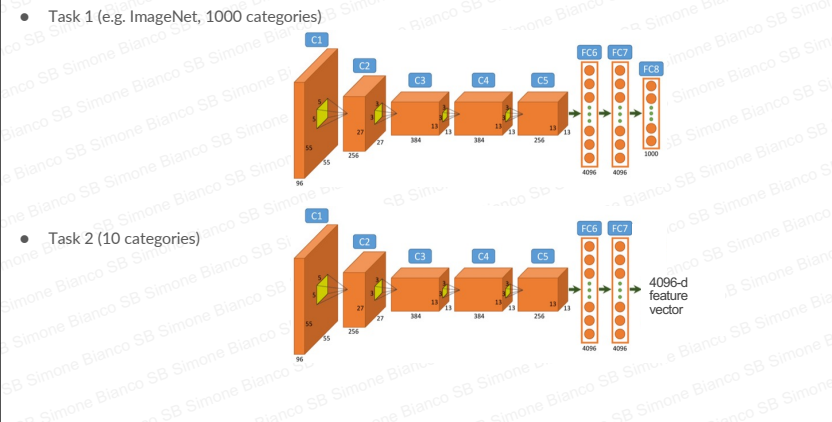
87

CNN as feature extractor



88

CNN as feature extractor



89

How to decide what type of TL you should perform?

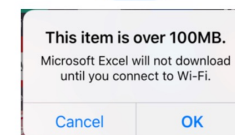
- New dataset is **small and similar** to original dataset:
 - Train a linear classifier on CNN features from higher layers
- New dataset is **large and similar** to original dataset
 - Fine-tune the CNN
- New dataset is **small but very different** from original dataset
 - Train a linear classifier on CNN features from lower layers
- New dataset is **large and very different** from original dataset
 - Train CNN from scratch or fine-tune it

90

Model Compression

The problem: If running DNN on Mobile...

APP developers suffer from the model size!



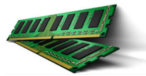
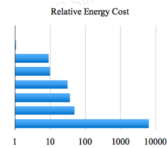
91

Model Compression

The problem: If running DNN on Mobile...

HW engineers suffer from the model size! (embedded systems, limited resources)

Operation	Energy [pJ]	Relative Cost
32 bit int ADD	0.1	1
32 bit float ADD	0.9	9
32 bit Register File	1	10
32 bit int MULT	3.1	31
32 bit float MULT	3.7	37
32 bit SRAM Cache	5	50
32 bit DRAM Memory	640	6400



92

Model Compression

The problem: If running DNN on the Cloud...

Network
Delay

Power
Budget

User
Privacy

Intelligent but Inefficient

93

Model Compression

Smaller Size
Compress Mobile App
Size by 35x-50x

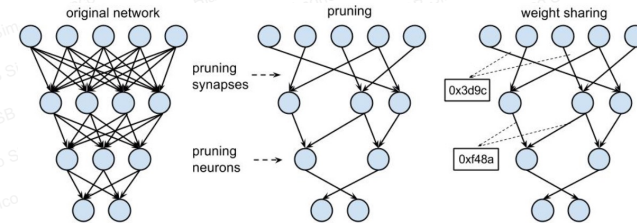
Accuracy
no loss of accuracy
improved accuracy

Speedup
make inference faster

- AlexNet: 35x, 240MB => 6.9MB
- VGG16: 49x 552MB => 11.3MB
- Both with no loss of accuracy on ImageNet
- Weights fits on-chip SRAM, taking 120x less energy than DRAM

94

Model Compression



95

Model Compression

- 50 trillions of synapses are generated in the human brain during the first few months of birth.
- 1 year old, peaked at 1000 trillions
- Pruning begins to occur.
- 10 years old, a child has nearly 500 trillions synapses
- This 'pruning' mechanism removes redundant connections in the brain.

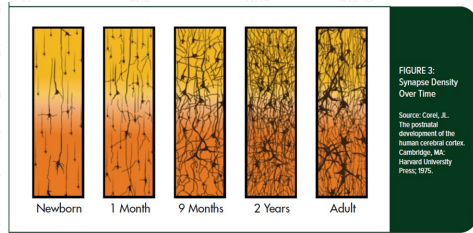


FIGURE 3:
Synapse Density
Over Time

Source: Corel, R.
The postnatal
development of the
human cerebral cortex.
Cambridge, MA:
Harvard University
Press, 1975.